



Cairo University

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Final Project MCQ

ELC 4028 – Artificial Neural Networks

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Q1. What does RAG stand for?

- A. Retrieval-Augmented Generation
- B. Random Attention Generator
- C. Retrieval Artificial Graph
- D. Recursive Augmented Graph

Correct Answer

A. Retrieval-Augmented Generation

RAG combines information retrieval systems with language generation models to improve answer quality and reduce hallucinations.

Q2. What is the main purpose of RAG systems?

- A. Compress neural networks
- B. Retrieve external knowledge before generation
- C. Replace transformers completely
- D. Reduce dataset size

Correct Answer

B. Retrieve external knowledge before generation

RAG systems first retrieve relevant documents and then use them to generate context-aware responses.

Q3. Which problem does RAG primarily help reduce?

- A. Overfitting
- B. Vanishing gradients
- C. Hallucination
- D. Image noise

Correct Answer

C. Hallucination

RAG reduces hallucinations by grounding responses using retrieved external information.

Q4. What are embeddings in RAG systems?

- A. Image compression algorithms
- B. Numerical vector representations of text
- C. Audio processing layers
- D. Database encryption methods

Correct Answer

B. Numerical vector representations of text

Embeddings convert text into vectors that capture semantic meaning and relationships.

Q5. Which database type is commonly used in RAG systems?

- A. Relational database
- B. Vector database
- C. Blockchain database
- D. Graphical database

Correct Answer

B. Vector database

Vector databases store embeddings and perform semantic similarity search efficiently.

Q6. Which similarity metric is commonly used for retrieval?

- A. Euclidean sorting
- B. Binary matching
- C. Cosine similarity
- D. Histogram equalization

Correct Answer

C. Cosine similarity

Cosine similarity measures the angular similarity between embedding vectors.

Q7. What happens during the retrieval phase in RAG?

- A. Model weights are retrained
- B. Relevant document chunks are searched
- C. Images are generated
- D. Audio signals are filtered

Correct Answer

B. Relevant document chunks are searched

The retrieval stage identifies semantically similar documents related to the user query.

Q8. What is chunking in RAG systems?

- A. Compressing neural weights
- B. Splitting documents into smaller sections
- C. Removing embeddings
- D. Encrypting text files

Correct Answer

B. Splitting documents into smaller sections

Large documents are divided into smaller chunks to improve retrieval accuracy.

Q9. Which of the following is an advantage of RAG compared to fine-tuning?

- A. Requires retraining for every update
- B. Easier knowledge updates
- C. Uses no embeddings
- D. Removes transformers completely

Correct Answer

B. Easier knowledge updates

RAG allows updating external documents without retraining the entire model.

Q10. Which ANN-related concept is strongly connected to RAG?

- A. Backpropagation only
- B. Embeddings and transformers
- C. Edge detection only
- D. Fourier transforms

Correct Answer

B. Embeddings and transformers

RAG relies heavily on transformer-based models and semantic vector embeddings.

Q11. What is semantic search?

- A. Searching by exact keyword only
- B. Searching based on meaning and context
- C. Searching image colors
- D. Searching file size

Correct Answer

B. Searching based on meaning and context

Semantic search retrieves information using contextual meaning rather than exact keyword matching.

Q12. Why are vector databases important in RAG systems?

- A. They store operating systems
- B. They perform fast similarity search
- C. They replace embeddings
- D. They generate neural networks

Correct Answer

B. They perform fast similarity search

Vector databases efficiently retrieve semantically similar embeddings during the retrieval phase.

Q13. What is the role of the embedding model in RAG systems?

- A. Generate images from text
- B. Convert text into vector representations
- C. Compress the database
- D. Remove irrelevant documents

Correct Answer

B. Convert text into vector representations

Embedding models transform text into numerical vectors that capture semantic meaning for similarity search.

Q14. Which component generates the final response in a RAG pipeline?

- A. Vector database
- B. Embedding layer
- C. Large Language Model (LLM)
- D. Data loader

Correct Answer

C. Large Language Model (LLM)

After retrieval, the LLM uses the retrieved context to generate the final answer.

Q15. What is one limitation of RAG systems?

- A. They cannot use transformers
- B. Poor retrieval can produce incorrect answers
- C. They completely eliminate latency
- D. They require no storage

Correct Answer

B. Poor retrieval can produce incorrect answers

If irrelevant or incorrect chunks are retrieved, the generated response quality decreases significantly.

Q16. Which of the following tools is commonly used for vector similarity search?

- A. FAISS
- B. Photoshop
- C. AutoCAD
- D. MySQL Workbench

Correct Answer

A. FAISS

FAISS is a popular library developed for efficient vector similarity search and retrieval.

Q17. Why is chunk size important in RAG systems?

- A. It controls internet speed
- B. It affects retrieval quality and context understanding
- C. It changes GPU temperature
- D. It removes embeddings

Correct Answer

B. It affects retrieval quality and context understanding

Very small chunks may lose context, while very large chunks may reduce retrieval precision.

Q18. Which type of search is performed in vector databases?

- A. Lexical search only
- B. Semantic similarity search
- C. Binary search only
- D. Pixel-based search

Correct Answer

B. Semantic similarity search

Vector databases retrieve information based on semantic meaning rather than exact word matching.

Q19. What is prompt augmentation in RAG systems?

- A. Reducing prompt size
- B. Adding retrieved context to the prompt
- C. Encrypting the prompt
- D. Removing user queries

Correct Answer

B. Adding retrieved context to the prompt
Retrieved documents are appended to the user query to improve answer accuracy.

Q20. Why are RAG systems considered useful for modern AI applications?

- A. They allow AI systems to use updated external knowledge
- B. They completely replace neural networks
- C. They remove the need for databases
- D. They eliminate all AI errors

Correct Answer

A. They allow AI systems to use updated external knowledge
RAG systems improve reliability by integrating external and dynamically updated information sources.